

Housing Reallocation and Demographic Transitions

Technical appendix for a two-generation OLG model

1 Environment and timing

Time is discrete, and one period is one generation. At date t , the economy contains Y_t young households and O_t old households. Each young household has a permanent type $i = (y_i, b_i)$ drawn from a probability distribution F , where y_i is current income and b_i is liquid wealth available at closing. Each old household exits at the end of the period. A child born at t generates $\nu > 0$ young households at $t + 1$ after survival and household formation. Children do not inherit their parent's individual financial state.

A young household consumes c , occupies h units of housing, and chooses completed fertility n . Its utility is

$$u_t^Y(c, h, n) = \log c + \alpha \log(h - \kappa n) + \vartheta_t \log n. \quad (1)$$

Each child uses $\chi > 0$ units of current goods and $\kappa > 0$ units of space. An old household consumes c^O , retains h^O units of housing, and leaves an estate e valued according to

$$u^O(c^O, h^O, e) = \log c^O + \gamma \log h^O + \omega_B \log e. \quad (2)$$

All choices lie in the log domain. In particular, $c, n, h - \kappa n, c^O, h^O$, and e are strictly positive.

The economy has a fixed housing stock $\bar{H}$. Each unit produces one unit of housing services in every period. Renters can occupy at most $\bar{h}^R$, while owners can occupy at most $\bar{h}^O > \bar{h}^R$. The house price is P_t . A claim to one unit of date- $t + 1$ goods costs $q \in (0, 1)$ at date t , and the corresponding gross return is $R = q^{-1}$.

A competitive housing intermediary buys a unit at t , pays the recurring property tax $\tau^p P_t$, rents the unit for r_t , and sells it at $t + 1$. Zero profit therefore requires

$$r_t = (1 + \tau^p)P_t - qP_{t+1}. \quad (3)$$

A young buyer finances the share $\phi \in (0, 1)$ of the purchase price with a one-period mortgage, posts $(1 - \phi)P_t h$ at closing, and repays $R\phi P_t h$ when old. A household sale at t realizes the per-unit tax

$$\ell_t = \tau^g [P_t - P_{t-1}]_+. \quad (4)$$

Housing retained until death receives a stepped-up basis and is liquidated by the estate without this household capital-gains liability. Intermediaries mark their basis to market. Old owners self-occupy retained housing and do not operate as landlords.

The cash flows imply two private user costs. A young buyer who sells the marginal unit when old pays

$$r_t^O = (1 + \tau^p)P_t - q(P_{t+1} - \ell_{t+1}) = r_t + q\ell_{t+1}. \quad (5)$$

An old incumbent who retains a marginal unit instead of selling it pays

$$r_t^I = (P_t - \ell_t) + \tau^p P_t - qP_{t+1} = r_t - \ell_t. \quad (6)$$

The buyer anticipates the tax on next-period resale, while the incumbent avoids the current realization tax by retaining the unit.

The timing within date t is: inherited states and P_{t-1} are observed; P_t, P_{t+1}, T_t , and T_{t+1} are correctly anticipated; young households choose tenure, housing, fertility, consumption, and saving; old households choose sales, retained housing, consumption, and estates; markets clear and taxes are rebated; the young become the next old and their children form the next young cohort.

2 Household problems and exact conditional solutions

2.1 Old renters and old incumbents

An old renter with financial wealth a solves

$$\begin{aligned} V_t^R(a) &= \max_{c^O, h^O, e} \log c^O + \gamma \log h^O + \omega_B \log e \\ \text{subject to } & c^O + qe + r_t h^O = a + T_t, \quad 0 < h^O \leq \bar{h}^R. \end{aligned} \quad (7)$$

An old incumbent enters with net financial wealth a and inherited housing H . Selling all housing produces $(P_t - \ell_t)H$. Expressing the estate in total goods, including retained housing, gives

$$\begin{aligned} V_t^O(a, H) &= \max_{c^O, h^O, e} \log c^O + \gamma \log h^O + \omega_B \log e \\ \text{subject to } & c^O + qe + (r_t - \ell_t)h^O = a + (P_t - \ell_t)H + T_t, \\ & 0 < h^O \leq H, \quad e \geq P_{t+1}h^O. \end{aligned} \quad (8)$$

The last inequality is the nonnegative liquid-estate restriction. It prevents the household from promising an estate smaller than the value of the retained house.

Let

$$A = 1 + \gamma + \omega_B, \quad \rho_t = r_t - \ell_t, \quad X_t = a + (P_t - \ell_t)H + T_t. \quad (9)$$

The old problems are strictly concave on their feasible sets. Their active sets therefore provide a complete characterization.

Lemma 1 (Old-age allocation). *Suppose $r_t > 0$, $\rho_t > 0$, and resources are sufficient for the log domain. If the renter size limit is slack, then*

$$c^O = \frac{a + T_t}{A}, \quad h^O = \frac{\gamma(a + T_t)}{Ar_t}, \quad e = \frac{\omega_B(a + T_t)}{Aq}. \quad (10)$$

If both incumbent constraints are slack, then

$$c^O = \frac{X_t}{A}, \quad h^O = \frac{\gamma X_t}{A\rho_t}, \quad e = \frac{\omega_B X_t}{Aq}. \quad (11)$$

The two slackness conditions are

$$\frac{\gamma X_t}{A\rho_t} < H, \quad \omega_B \rho_t > \gamma q P_{t+1}. \quad (12)$$

Proof. For a generic budget $c^O + qe + p_h h^O = X$, the three first-order conditions imply $p_h h^O = \gamma c^O$ and $qe = \omega_B c^O$. The budget then gives $Ac^O = X$. Substituting $p_h = r_t$ for a renter and $p_h = \rho_t$ for an incumbent gives equations (10) and (11). Substituting the incumbent solution into $h^O < H$ and $e > P_{t+1}h^O$ gives equation (12). $\square$

For completeness, the remaining incumbent active sets also have closed forms. If only estate composition binds, define $d_t = \rho_t + qP_{t+1}$; then

$$c^O = \frac{X_t}{A}, \quad h^O = \frac{(\gamma + \omega_B)X_t}{Ad_t}, \quad e = P_{t+1}h^O. \quad (13)$$

If only retention binds, let $Z_t = X_t - \rho_t H$; then

$$h^O = H, \quad c^O = \frac{Z_t}{1 + \omega_B}, \quad e = \frac{\omega_B Z_t}{(1 + \omega_B)q}, \quad (14)$$

provided the estate inequality remains slack. If both constraints bind, $h^O = H$, $e = P_{t+1}H$, and

$$c^O = X_t - (\rho_t + qP_{t+1})H. \quad (15)$$

The old renter has the analogous size-cap solution: set $h^O = \bar{h}^R$ and allocate the remaining resources between c^O and e in shares $1 : (\omega_B)$. The appropriate row is selected by feasibility and complementary slackness. For example, the estate-binding row requires $\gamma q P_{t+1} \geq \omega_B \rho_t$; the retention-only row requires

$$\omega_B Z_t > (1 + \omega_B)qP_{t+1}H, \quad \gamma Z_t \geq (1 + \omega_B)\rho_t H. \quad (16)$$

When both constraints bind, the two multipliers are nonnegative exactly when

$$\frac{q}{c^O} \geq \frac{\omega_B}{P_{t+1}H}, \quad \frac{\gamma + \omega_B}{H} \geq \frac{\rho_t + qP_{t+1}}{c^O}. \quad (17)$$

In the fully interior incumbent region, the indirect value and its wealth envelope are

$$V_t^O(a, H) = A \log X_t - A \log A + \gamma \log \gamma - \gamma \log \rho_t + \omega_B \log \omega_B - \omega_B \log q, \quad (18)$$

$$V_{a,t}^O = \frac{A}{X_t} = \frac{1}{c_t^O}, \quad V_{H,t}^O = (P_t - \ell_t)V_{a,t}^O. \quad (19)$$

These envelopes are the link between the young buyer's current problem and her old-age resources.

2.2 Young renters and young owners

Conditional on renting, a young household solves

$$\begin{aligned} W_t^R(i) &= \max_{c,h,n,a'} u_t^Y(c, h, n) + \beta V_{t+1}^R(Ra') \\ \text{subject to} \quad &c + \chi n + a' + r_t h = y_i + b_i + T_t, \\ &0 < h \leq \bar{h}^R, \quad a' \geq 0. \end{aligned} \quad (20)$$

Conditional on ownership, it solves

$$\begin{aligned} W_t^O(i) &= \max_{c,h,n,a'} u_t^Y(c, h, n) + \beta V_{t+1}^O(a_{t+1}, h) \\ \text{subject to} \quad &c + \chi n + a' + [(1 - \phi) + \tau^p]P_t h = y_i + b_i + T_t, \\ &a_{t+1} = Ra' - R\phi P_t h, \\ &(1 - \phi)P_t h \leq b_i, \quad 0 < h \leq \bar{h}^O, \quad a' \geq 0. \end{aligned} \quad (21)$$

Liquid wealth b_i , rather than income received after closing, is pledgeable for the down payment.

The first-order conditions also characterize cases in which saving or a housing limit binds. Let λ_t^m be the multiplier on the current budget, σ_t^m the multiplier on $a' \geq 0$, η_t^m the multiplier on the tenure-specific size limit, and μ_t the owner down-payment multiplier. The common consumption and fertility conditions are

$$\lambda_t^m = \frac{1}{c_t^m}, \quad (22)$$

$$\frac{\vartheta_t}{n_t^m} = \lambda_t^m \chi + \frac{\alpha \kappa}{h_t^m - \kappa n_t^m}. \quad (23)$$

The saving conditions are

$$\lambda_t^R = \beta R V_{a,t+1}^R + \sigma_t^R, \quad (24)$$

$$\lambda_t^O = \beta R V_{a,t+1}^O + \sigma_t^O, \quad (25)$$

with $\sigma_t^m a_t' = 0$. When saving and the future old constraints are slack, the housing conditions reduce to

$$\frac{\alpha c_t^m}{h_t^m - \kappa n_t^m} = r_t^m + \zeta_t^m, \quad r_t^R = r_t, \quad r_t^O = r_t + q\ell_{t+1}, \quad (26)$$

where the current-goods access values are

$$\zeta_t^R = \frac{\eta_t^R}{\lambda_t^R}, \quad \zeta_t^O = \underbrace{\frac{\mu_t}{\lambda_t^O}(1 - \phi)P_t}_{\zeta_t^{O,F}} + \underbrace{\frac{\eta_t^O}{\lambda_t^O}}_{\zeta_t^{O,S}}. \quad (27)$$

Thus $\zeta_t^{O,F}$ is the financing component and $\zeta_t^{O,S}$ is the owner-size component. If both limits bind, choices identify their sum but not the two multipliers separately.

2.3 Lifetime reduction and unconstrained policies

Suppose young saving and the future old allocation are interior. The saving Euler equation and equation (11) imply

$$\frac{1}{c_t} = \beta R \frac{1}{c_{t+1}^O}, \quad c_{t+1}^O = \beta R c_t, \quad qX_{t+1} = kc_t, \quad k \equiv \beta A. \quad (28)$$

For an owner,

$$X_{t+1} = Ra_t' - R\phi P_t h_t + (P_{t+1} - \ell_{t+1})h_t + T_{t+1}. \quad (29)$$

Combining current and discounted old resources gives the lifetime identity

$$(1 + k)c_t + \chi n_t + r_t^m h_t = \mathcal{W}_{it}, \quad \mathcal{W}_{it} = y_i + b_i + T_t + qT_{t+1}. \quad (30)$$

The renter identity is the same with $r_t^m = r_t$; the owner identity uses $r_t^m = r_t + q\ell_{t+1}$. This is a private lifetime budget, not an aggregate resource constraint.

Proposition 1 (Unconstrained conditional policies). *Let*

$$B_t = 1 + k + \alpha + \vartheta_t. \quad (31)$$

When saving, housing limits, and future old constraints are all slack, the conditional young policies under tenure m are

$$c_t^m = \frac{\mathcal{W}_{it}}{B_t}, \quad (32)$$

$$n_t^m = \frac{\vartheta_t \mathcal{W}_{it}}{B_t(\chi + \kappa r_t^m)}, \quad (33)$$

$$h_t^m - \kappa n_t^m = \frac{\alpha \mathcal{W}_{it}}{B_t r_t^m}, \quad (34)$$

$$h_t^m = \frac{\alpha \mathcal{W}_{it}}{B_t r_t^m} + \frac{\kappa \vartheta_t \mathcal{W}_{it}}{B_t(\chi + \kappa r_t^m)}. \quad (35)$$

The associated saving policies are

$$a_t'^R = kc_t^R - qT_{t+1}, \quad (36)$$

$$a_t'^O = kc_t^O - qT_{t+1} - [q(P_{t+1} - \ell_{t+1}) - \phi P_t]h_t^O. \quad (37)$$

Proof. Let $s = h - \kappa n$ denote family space. The unconstrained housing condition gives $s = \alpha c / r_t^m$. Combining the fertility condition with this result gives $n = \vartheta_t c / (\chi + \kappa r_t^m)$. Substituting $h = s + \kappa n$ in equation (30) gives $\mathcal{W}_{it} = B_t c$. Equations (36) and (37) follow from $qX_{t+1} = kc_t$. $\square$

The effective housing limits are

$$\hat{h}_t^R = \bar{h}^R, \quad \hat{h}_{it}^O = \min \left\{ \bar{h}^O, \frac{b_i}{(1 - \phi)P_t} \right\}. \quad (38)$$

A limit binds exactly when the unconstrained policy in equation (35) exceeds the relevant $\hat{h}$. When it binds, $h = \hat{h}$ and fertility is the unique root of

$$F(n, \hat{h}) \equiv \frac{\vartheta_t}{n} - \frac{\alpha \kappa}{\hat{h} - \kappa n} - \frac{(1 + k)\chi}{\mathcal{W}_{it} - r_t^m \hat{h} - \chi n} = 0. \quad (39)$$

The root lies in

$$0 < n < \min \left\{ \frac{\hat{h}}{\kappa}, \frac{\mathcal{W}_{it} - r_t^m \hat{h}}{\chi} \right\}. \quad (40)$$

Since $F_n < 0$, the root is unique. Consumption and saving then follow from equations (30), (36), and (37). If $a' = 0$, the lifetime reduction no longer applies; equations (22)–(25) and the binding resource constraint uniquely characterize the conditional solution.

The household chooses $m_t(i) \in \{R, O\}$ by comparing the two conditional values. Conditional problems are strictly concave, but the tenure decision is discrete. The aggregate analysis below uses the following standard regularity condition.

Assumption 1 (Tenure regularity). The type distribution is atomless at tenure indifference: for every equilibrium price and transfer vector, $F\{i : W_t^R(i) = W_t^O(i)\} = 0$. Conditional policies are unique, and parameters remain in a compact region in which resources and relevant user costs are strictly positive.

Under Assumption 1, individual tenure switches do not create jumps in aggregate choice maps. A continuous idiosyncratic tenure taste is one primitive way to satisfy the condition, but it is not needed for the household derivations.

3 Aggregation and positive steady states

3.1 States, distributions, and government accounts

Let G_t be the normalized distribution of old financial wealth, inherited housing, and tenure. It is generated by the preceding young policies:

$$G_{t+1} = \text{Law}_{i \sim F} (Ra'_t(i) - \mathbf{1}\{m_t(i) = O\}R\phi P_t h_t(i), \mathbf{1}\{m_t(i) = O\}h_t(i), m_t(i)). \quad (41)$$

For every bounded measurable test function ψ , this statement means

$$\int \psi(a, H, m) dG_{t+1} = \int \psi(Ra'_t(i) - \mathbf{1}\{m_t(i) = O\}R\phi P_t h_t(i), \mathbf{1}\{m_t(i) = O\}h_t(i), m_t(i)) dF(i). \quad (42)$$

Because children do not inherit individual states, this is a one-generation pushforward rather than an intergenerational wealth fixed point.

Let $\bar{n}_t$, $\bar{h}_t^Y$, and $\bar{h}_t^O$ be average fertility, young housing, and old housing. Let

$$\bar{s}_t^O = \int \mathbf{1}\{m = O\} [H - h_t^O(a, H, m)] dG_t \quad (43)$$

be average taxable old-owner sales, assigning zero to renters. Cohorts and housing satisfy

$$Y_{t+1} = \nu \bar{n}_t Y_t, \quad O_{t+1} = Y_t, \quad Y_t \bar{h}_t^Y + O_t \bar{h}_t^O = \bar{H}. \quad (44)$$

The government rebates property- and gains-tax revenue equally across living households:

$$(Y_t + O_t)T_t = \tau^p P_t \bar{H} + \ell_t O_t \bar{s}_t^O. \quad (45)$$

Property taxation applies to the fixed occupied stock. The gains-tax base is the housing actually sold by old owners.

3.2 A two-equation characterization of a positive steady state

Fix a constant fertility preference ϑ . Let $\mathcal{N}(P, T; \vartheta)$ denote average young fertility and let $\mathcal{H}^Y(P, T; \vartheta)$ and $\mathcal{H}^O(P, T; \vartheta)$ denote stationary housing per young and old household after solving the household problems and applying the pushforward in equation (41). Write

$$\mathcal{H}(P, T; \vartheta) = \mathcal{H}^Y(P, T; \vartheta) + \mathcal{H}^O(P, T; \vartheta). \quad (46)$$

Definition 1 (Positive steady state). A positive steady state is a constant equilibrium with $Y^* = O^* > 0$, constant P^* and T^* , stationary policies and G^* , and all household, housing, cohort, and government conditions satisfied.

In any steady state, $P_t = P_{t-1} = P_{t+1} = P^*$, and therefore

$$\ell^* = 0, \quad r^* = \delta P^*, \quad \delta \equiv 1 + \tau^p - q > 0. \quad (47)$$

The absence of real capital gains at a stationary endpoint follows from the tax base itself; it does not require a separate assumption.

Proposition 2 (Reduced steady-state system). *Every positive steady state is characterized by a root $(P^*, T^*) \in \mathbb{R}_{++}^2$ of*

$$F_1(P, T; \vartheta) \equiv \nu \mathcal{N}(P, T; \vartheta) - 1 = 0, \quad (48)$$

$$F_2(P, T; \vartheta) \equiv T - \frac{\tau^p P}{2} \mathcal{H}(P, T; \vartheta) = 0. \quad (49)$$

Given the root, the population scale and old distribution are

$$Y^* = O^* = \frac{\bar{H}}{\mathcal{H}(P^*, T^*; \vartheta)}, \quad G^* = \Gamma(P^*, T^*; \vartheta). \quad (50)$$

Consequently, every positive steady state has average fertility $\mathcal{N}(P^, T^*; \vartheta) = 1/\nu$.*

Proof. Stationarity of $O_{t+1} = Y_t$ gives $O^* = Y^*$. Stationarity of the young cohort then gives $1 = \nu \mathcal{N}$. Housing clearing gives $\bar{H} = Y^* \mathcal{H}$. Since $\ell^* = 0$, the government budget gives $2Y^* T^* = \tau^p P^* \bar{H}$. Eliminating Y^* using housing clearing produces equation (49); reversing these steps establishes sufficiency. $\square$

The positive steady state is therefore a property of primitives and policy, not a normalization. It need not exist for arbitrary parameters. The next result provides transparent sufficient conditions.

Proposition 3 (Existence and local uniqueness). *Suppose Assumption 1 holds on the rectangle $\mathcal{D} = [\underline{P}, \overline{P}] \times [\underline{T}, \overline{T}] \subset \mathbb{R}_{++}^2$. Suppose further that:*

1. *the fiscal map $\Psi(P, T) = (\tau^p P/2)\mathcal{H}(P, T; \vartheta)$ maps $[\underline{T}, \overline{T}]$ into itself for every P and is a uniform contraction in T ;*
2. *its unique fixed point $T = \mathcal{T}(P)$ is interior and continuous; and*
3. *the replacement residual changes sign:*

$$F_1(\underline{P}, \mathcal{T}(\underline{P}); \vartheta)F_1(\overline{P}, \mathcal{T}(\overline{P}); \vartheta) < 0. \quad (51)$$

Then a positive steady state exists in $\mathcal{D}$. It is locally unique if the Jacobian

$$J = \begin{bmatrix} \nu\mathcal{N}_P & \nu\mathcal{N}_T \\ -\frac{\tau^p}{2}(\mathcal{H} + P\mathcal{H}_P) & 1 - \frac{\tau^p P}{2}\mathcal{H}_T \end{bmatrix}_{(P^*, T^*; \vartheta)} \quad (52)$$

is nonsingular.

Proof. The contraction theorem gives a unique continuous fiscal locus $T = \mathcal{T}(P)$. The intermediate value theorem applied to $F_1(P, \mathcal{T}(P); \vartheta)$ gives existence. Nonsingularity of J and the implicit-function theorem give local uniqueness. $\square$

When the Jacobian is nonsingular, a change in the fertility preference has the local comparative static

$$\begin{bmatrix} P_\vartheta^* \\ T_\vartheta^* \end{bmatrix} = -J^{-1} \begin{bmatrix} \nu\mathcal{N}_\vartheta \\ -\frac{\tau^p P^*}{2}\mathcal{H}_\vartheta \end{bmatrix}. \quad (53)$$

The population-scale response then follows from

$$\frac{d \log Y^*}{d\vartheta} = -\frac{\mathcal{H}_P P_\vartheta^* + \mathcal{H}_T T_\vartheta^* + \mathcal{H}_\vartheta}{\mathcal{H}}. \quad (54)$$

Equations (53)–(54) show why replacement fertility can be invariant across endpoints while prices, transfers, individual policies, the old-state distribution, and population scale differ.

4 Perfect-foresight transition equilibrium

Suppose $\vartheta_t = \vartheta^0$ for $t < 0$ and $\vartheta_t = \vartheta^1$ for $t \geq 0$, and that both values admit positive steady states $\mathcal{E}^0$ and $\mathcal{E}^1$. The predetermined aggregate state is

$$S_t = (Y_t, O_t, G_t, P_{t-1}). \quad (55)$$

The lagged price records the basis of the old owners who may sell at t . Prices and transfers dated t are not predetermined.

Definition 2 (Transition equilibrium). Given $S_0 = (Y^0, Y^0, G^0, P^0)$, a perfect-foresight transition equilibrium is a sequence $\{P_t, T_t, S_t\}_{t \geq 0}$ and household policies such that:

1. every old and young household solves its problem under the correctly anticipated sequence;
2. G_{t+1} is the pushforward in equation (41);
3. housing clears and the government budget balances at every date;
4. cohorts evolve according to equation (44);
5. ℓ_t and r_t satisfy equations (4) and (3); and
6. (P_t, T_t, S_t) converges to $\mathcal{E}^1$.

This definition initializes one inherited state, not two shocks. The date-zero old are those generated by the old steady state, and their basis is $P_{-1} = P^0$. The permanent preference change is the only exogenous innovation.

To display the equilibrium operator, let household solutions under a candidate perfect-foresight sequence generate $\bar{h}_t^Y$, $\bar{h}_t^O$, $\bar{n}_t$, and $\bar{s}_t^O$. At each date the two equilibrium residuals are

$$\mathcal{R}_t^H = Y_t \bar{h}_t^Y + O_t \bar{h}_t^O - \bar{H}, \quad (56)$$

$$\mathcal{R}_t^G = (Y_t + O_t)T_t - \tau^p P_t \bar{H} - \ell_t O_t \bar{s}_t^O. \quad (57)$$

Together with the forward laws for S_{t+1} , a transition is a root of the sequence operator $\mathcal{R} = \{\mathcal{R}_t^H, \mathcal{R}_t^G\}_{t \geq 0}$.

The capital-gains rule can equivalently be written as the complementarity system

$$\ell_t \geq 0, \quad \ell_t \geq \tau^g(P_t - P_{t-1}), \quad \ell_t [\ell_t - \tau^g(P_t - P_{t-1})] = 0. \quad (58)$$

The steady state lies at the kink. Ordinary derivatives are therefore regime-specific: the appreciating regime has derivative τ^g , while the non-appreciating regime has derivative zero.

Proposition 4 (Finite-horizon existence and local determinacy). *Fix a terminal date J and impose the terminal continuation $\mathcal{E}^1$ from $J + 1$ onward. Suppose aggregate policies are continuous, the induced state law remains in a compact invariant set, and the mapping from a candidate vector $(P_0, \dots, P_J, T_0, \dots, T_J)$ to the vector that clears equations (56)–(57) is a continuous self-map of a compact convex set. Then a finite-horizon transition exists. Within a fixed tenure, constraint, and appreciation regime, it is locally unique if the Jacobian of the stacked residual vector with respect to the price-transfer vector is nonsingular.*

Proof. Brouwer's theorem gives a fixed point of the continuous clearing map. Within a fixed active set, the residual vector is differentiable. A nonsingular stacked Jacobian and the implicit-function theorem then give local uniqueness. $\square$

The proposition states sufficient, verifiable conditions; it does not assert that every parameterization has a transition. At an appreciation kink one must either verify the candidate active regime, solve the complementarity problem directly, or use a generalized-Jacobian argument. For an infinite horizon, a sequence of finite-horizon solutions converges to a transition if it is uniformly bounded and has a convergent subsequence whose limit satisfies the terminal condition and equilibrium equations.

Corollary 1 (One shock can produce gains for many cohorts). *Following the single permanent preference shock, if the equilibrium price satisfies $P_s > P_{s-1}$ for $s = t, \dots, t + J$, then $\ell_s > 0$ on all $J + 1$ dates. A two-period household life does not restrict capital gains to one date.*

Proof. Equation (4) gives $\ell_s = \tau^g(P_s - P_{s-1}) > 0$ on each appreciating date. Two-period lives determine whose basis is P_{s-1} ; they do not determine the number of dates on which the equilibrium price can appreciate. $\square$

5 Intergenerational housing allocation

5.1 Marginal values

Consider a young owner i whose down-payment constraint binds, whose owner-size limit is slack, and whose saving and next-period old constraints are slack. Consider an old incumbent j whose retention and estate-composition constraints are slack and for whom $r_t > \ell_t$. Their marginal values of one unit of current housing services, in date- t goods, are

$$MV_{it}^Y = r_t + q\ell_{t+1} + \zeta_{it}^{O,F}, \quad MV_{jt}^O = r_t - \ell_t. \quad (59)$$

Lemma 2 (Transition valuation gap). *Under the stated active sets,*

$$MV_{it}^Y - MV_{jt}^O = \zeta_{it}^{O,F} + \ell_t + q\ell_{t+1}. \quad (60)$$

Proof. Subtract the incumbent user cost in equation (6) from the buyer's constrained marginal value in equation (26). The common rental user cost cancels. $\square$

The three terms have distinct sources. The financing wedge is the value of pledgeable liquidity at closing. The current gains tax lowers the incumbent's private cost of retention. The discounted future gains tax raises the buyer's cost of acquiring a unit that will be sold when old.

5.2 Steady-state compensated reallocation

The welfare comparison first holds fertility, cohort masses, prices, tenure, and all other household choices fixed. It changes only a marginal housing unit and balanced transfers between the two living households.

Definition 3 (Local compensated housing test). An allocation passes the local compensated housing test if no infinitesimal transfer of title and occupancy from an old incumbent to a current young owner, together with feasible balanced goods transfers, can leave both weakly better off and one strictly better off. The current young household's lifetime utility includes her own old age.

Let $K_{ij} \geq 0$ be the real date- t resource cost per unit of executing and servicing the reallocation.

Proposition 5 (Steady-state reallocation). *At a positive steady state, the goods-equivalent surplus from transferring a marginal unit from old incumbent j to financing-constrained young owner i is*

$$\mathcal{S}_{ij}^* = \zeta_i^{O,F} - K_{ij}. \quad (61)$$

If $\mathcal{S}_{ij}^ > 0$, the steady-state allocation fails the local compensated housing test.*

Proof. At a steady state, $\ell_t = \ell_{t+1} = 0$, so the valuation gap is $\zeta_i^{O,F}$. Transfer $\varepsilon > 0$ units of title and occupancy from j to i . To preserve the buyer's binding closing constraint, lend

$$L = (1 - \phi)P^*\varepsilon \quad (62)$$

immediately before closing and require repayment RL when the buyer is old. The bridge has zero present value. Let λ_i and μ_i be the buyer's budget and down-payment multipliers. Its contribution to the buyer's Lagrangian is

$$(\lambda_i + \mu_i)L - \beta V_{a,i,t+1}^O RL = \mu_i L = \lambda_i \zeta_i^{O,F} \varepsilon, \quad (63)$$

where $\lambda_i = \beta RV_{a,i,t+1}^O$ is the saving Euler equation. Thus the bridge relaxes the closing constraint without creating net financial wealth.

The incumbent sells the same marginal unit. At an interior retention choice, net sale proceeds and the induced estate adjustment compensate the first-order loss of housing services because $u_h^O/u_c^O = r^*$. The fixed property-tax base is unchanged, and no gains tax is realized at a constant price. After paying $K_{ij}\varepsilon$, the coalition has $(\zeta_i^{O,F} - K_{ij})\varepsilon$ in date- t goods. If this amount is positive, a sufficiently small transfer can be divided so that neither household loses and one gains strictly. $\square$

The proposition is a sufficient local result. A nonpositive surplus for one pair does not establish global efficiency: another pair, active set, or reallocation instrument may still generate gains.

5.3 A two-date transition result

Along a transition, the tax terms in equation (60) change government rebates. A Pareto statement therefore requires an explicit two-date transfer technology rather than only a comparison of private marginal values.

Assumption 2 (Two-date compensation technology). A reallocation authority can execute a taxable marginal sale, issue the zero-present-value pre-closing bridge in equation (62), trade one-period claims at price q , require the transferred marginal title to be resold at $t + 1$, make balanced lump-sum side transfers at t and $t + 1$, reclaim only incremental tax rebates, and pay real cost K_{ijt} in date- t goods.

Proposition 6 (Transition reallocation). *Under Assumption 2 and the active sets used in equation (59), the present-value surplus from a marginal transfer is*

$$\mathcal{S}_{ijt} = \zeta_{it}^{O,F} + \ell_t + q\ell_{t+1} - K_{ijt}. \quad (64)$$

If $\mathcal{S}_{ijt} > 0$, the transfer supports a local two-date constrained Pareto improvement.

Proof. Transfer ε units. The bridge loan produces the financing gain $\zeta_{it}^{O,F}\varepsilon$ by the same Euler-equation cancellation as in equation (63). The old incumbent is compensated to first order by the sale proceeds and estate adjustment implied by her interior retention condition.

The extra sale realizes $\ell_t\varepsilon$ in date- t tax revenue. The required next-period resale is feasible without changing the buyer's active set because her retention constraint is slack; it realizes $\ell_{t+1}\varepsilon$ at $t + 1$. If $N_s = Y_s + O_s$, the equal-rebate rule implies

$$N_t dT_t = \ell_t \varepsilon, \quad N_{t+1} dT_{t+1} = \ell_{t+1} \varepsilon. \quad (65)$$

The authority reclaims exactly each household's incremental rebate, so every other household remains at its baseline allocation. This yields $\ell_t\varepsilon + q\ell_{t+1}\varepsilon$ in date- t present value. Charging the buyer no more than its bridge gain, paying $K_{ijt}\varepsilon$, and distributing the strictly positive residual leaves every affected household weakly better off and at least one strictly better off. Differentiability and a strict first-order inequality imply a genuine improvement for sufficiently small ε . $\square$

The compensation experiment fixes fertility and population. If a policy changes the number or identity of future households, ordinary Pareto comparisons do not by themselves rank the two populations. One must state an endogenous-population welfare criterion, such as the P - or A -efficiency concepts developed by [Golosov, Jones, and Tertilt(2007)]; see also [Pérez-Nievas et al.(2019)]. No such criterion is needed for Propositions 5 and 6 because every household whose resources change is held weakly above its baseline and cohort masses are fixed.

6 Housing access and fertility

The reallocation result concerns willingness to pay for a fixed housing unit. The fertility result instead asks how an access limit changes a household's optimal n while tenure, prices, lifetime wealth, and the active set are fixed.

Suppose $h = \hat{h}$ binds and fertility remains interior. Define

$$D = \mathcal{W}_{it} - r_t^m \hat{h} - \chi n, \quad S = \hat{h} - \kappa n, \quad c = \frac{D}{1+k}. \quad (66)$$

Implicit differentiation of equation (39) gives the exact partial-equilibrium derivative

$$\frac{dn}{d\hat{h}} = \frac{\frac{\alpha\kappa}{S^2} - \frac{(1+k)\chi r_t^m}{D^2}}{\frac{\vartheta_t}{n^2} + \frac{(1+k)\chi^2}{D^2} + \frac{\alpha\kappa^2}{S^2}}. \quad (67)$$

The denominator is strictly positive. More family space raises the return to fertility, represented by the first numerator term; the goods cost of occupying additional housing lowers disposable consumption, represented by the second.

At a constrained allocation, equation (26) implies

$$\frac{D}{S} = \frac{(1+k)(r_t^m + \zeta_t^m)}{\alpha}. \quad (68)$$

For $r_t^m > 0$, substitution gives the exact sign condition

$$\frac{dn}{d\hat{h}} \geq 0 \iff \chi \leq \frac{(1+k)\kappa(r_t^m + \zeta_t^m)^2}{\alpha r_t^m}. \quad (69)$$

A stronger condition stated only in primitives and the user cost is

$$\chi \leq \frac{(1+k)\kappa r_t^m}{\alpha}. \quad (70)$$

The sign is therefore economic, not automatic: a child must be sufficiently space intensive relative to its goods cost.

When the down-payment limit is the active owner constraint,

$$\hat{h}_{it}^O = \frac{b_i}{(1-\phi)P_t}, \quad (71)$$

and its mechanical derivatives are

$$\frac{\partial \hat{h}^O}{\partial b_i} = \frac{1}{(1-\phi)P_t}, \quad \frac{\partial \hat{h}^O}{\partial \phi} = \frac{b_i}{(1-\phi)^2 P_t}, \quad \frac{\partial \hat{h}^O}{\partial P_t} = -\frac{\hat{h}^O}{P_t}. \quad (72)$$

For a reform that changes only the pledgeable cap locally, $\partial n / \partial z = (dn/d\hat{h})(\partial \hat{h} / \partial z)$. A funded policy or a general-equilibrium change also changes $\mathcal{W}$, r_t^m , rebates, and possibly tenure; those additional channels must be included before assigning an aggregate sign.

Table 1: Parameters of the worked equilibrium

Object	Parameter	Value	Object	Parameter	Value
Discount price	q	0.50	Mortgage share	ϕ	0.80
Property tax	τ^p	0.05	Gains-tax rate	τ^g	0.20
Space weight	α	0.40	Space per child	κ	0.50
Goods per child	χ	0.15	Young discount	β	0.40
Old housing weight	γ	0.30	Estate weight	ω_B	0.40
Income	y	1.00	Liquid wealth	b	0.06
Household formation	ν	2.00	Housing stock	$\bar{H}$	2.00

7 A solved equilibrium path

This section gives a reproducible equilibrium example of the preceding equations. It uses the owner-only restriction of the model, a binding down-payment limit, interior saving, and interior old choices. The example is a constructive existence check for this restricted parameterization rather than an empirical calibration.

At date zero, ϑ rises permanently from 0.35 to 0.50. Solving equations (48)–(49) gives the two endpoints in Table 2.

Table 2: Steady-state endpoints of the worked equilibrium

Object	Initial steady state	Final steady state
Fertility preference ϑ	0.3500	0.5000
Average fertility $\bar{n}$	0.5000	0.5000
House price P	0.3999	0.5599
Per-household rebate T	0.0130	0.0130
Young housing h^Y	0.7503	0.5358
Old housing h^O	0.5453	0.3894
Young and old cohort mass $Y = O$	1.5438	2.1617

Average fertility equals $1/\nu = 0.5$ at both endpoints, as required by replacement. The endpoints nevertheless differ in prices, individual housing, and population scale. On impact, fertility rises above replacement. The young cohort consequently expands, and the house price appreciates while the fixed stock is reallocated across changing cohort masses.

The finite-horizon root uses the final steady state as the continuation value after date 24. The maximum absolute housing or government residual is 3.2×10^{-10} , and the maximum terminal-state gap is 2.4×10^{-10} . Along the path, the minimum old sale is 0.145, the minimum incumbent user cost is 0.206, the minimum estate-composition slack is 0.029, minimum saving is 0.408, and the minimum difference between unconstrained owner housing and the down-payment cap is 0.348. The active sets used in the analytical derivation therefore hold strictly in the computed path. Replacing the terminal horizon 24 with horizon 20 changes the first nine dates by at most 1.4×10^{-15} , and the analytical fertility derivative in equation (67) matches a centered finite-difference derivative to within 2.0×10^{-10} .

The example establishes the narrow transition point mechanically. A single permanent preference change moves fertility on impact, cohort sizes with a lag, and prices over several dates. Capital gains are a transition object and vanish at both steady-state endpoints.

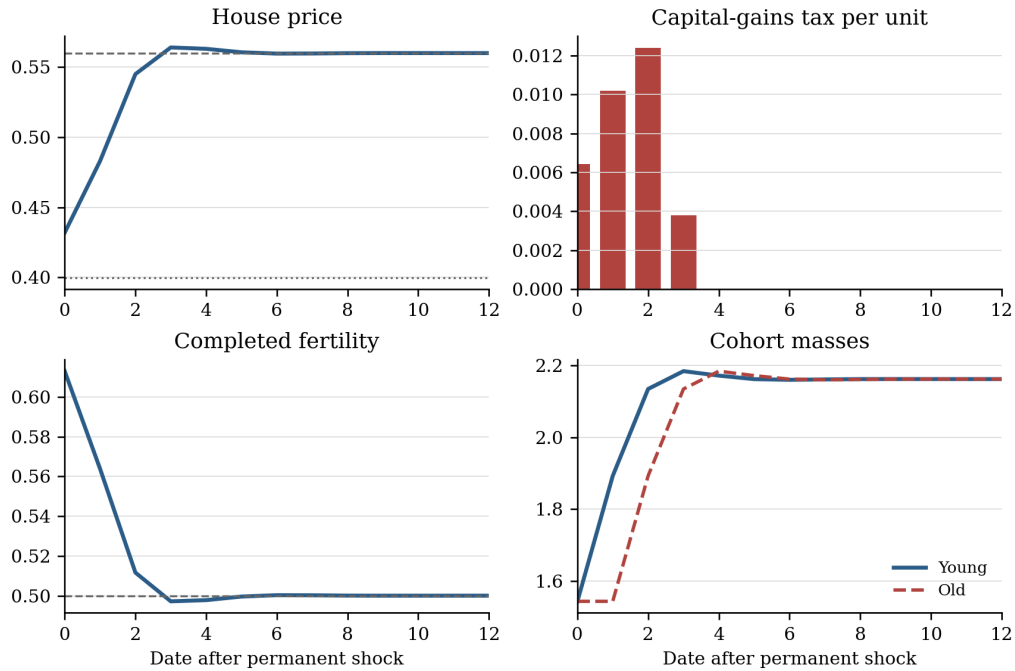


Figure 1: Perfect-foresight response to one permanent increase in the fertility preference. Dotted and dashed lines in the price panel denote the initial and final steady-state prices. The gains tax is positive on each of the first four dates because the equilibrium price continues to appreciate. Fertility moves on impact, cohort masses respond with the OLG lag, and both return to a positive steady state.

8 Result map

The model delivers four separate conclusions.

1. A positive steady state, when it exists, is a root of the two equations in (P, T) given by equations (48)–(49). Replacement fixes average fertility but not prices, policies, distributions, or population scale.
2. A single permanent shock can generate a multi-date price transition and capital gains for several successive old cohorts. A second shock is not required.
3. At a steady state, a financing wedge supports a local compensated housing reallocation when $\zeta_i^{O,F} > K_{ij}$. Along a transition, the analogous two-date surplus additionally contains $\ell_t + q\ell_{t+1}$, provided the rebate and transfer ledger in Assumption 2 is available.
4. Relaxing a housing-access limit raises fertility under the exact condition in equation (69). This response is distinct from the Pareto comparison and is not, by itself, a general-equilibrium welfare result.

References

- [Goloso, Jones, and Tertilt(2007)] Goloso, Mikhail, Larry E. Jones, and Michèle Tertilt. 2007. “Efficiency with Endogenous Population Growth.” *Econometrica* 75(4): 1039–1071.
- [Pérez-Nievas et al.(2019)] Pérez-Nievas, Mikel, José I. Conde-Ruiz, and Eduardo L. Giménez. 2019. “Efficiency and Endogenous Fertility.” *Theoretical Economics* 14(2): 475–512.